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Scientists just overturned a century-old physics assumption

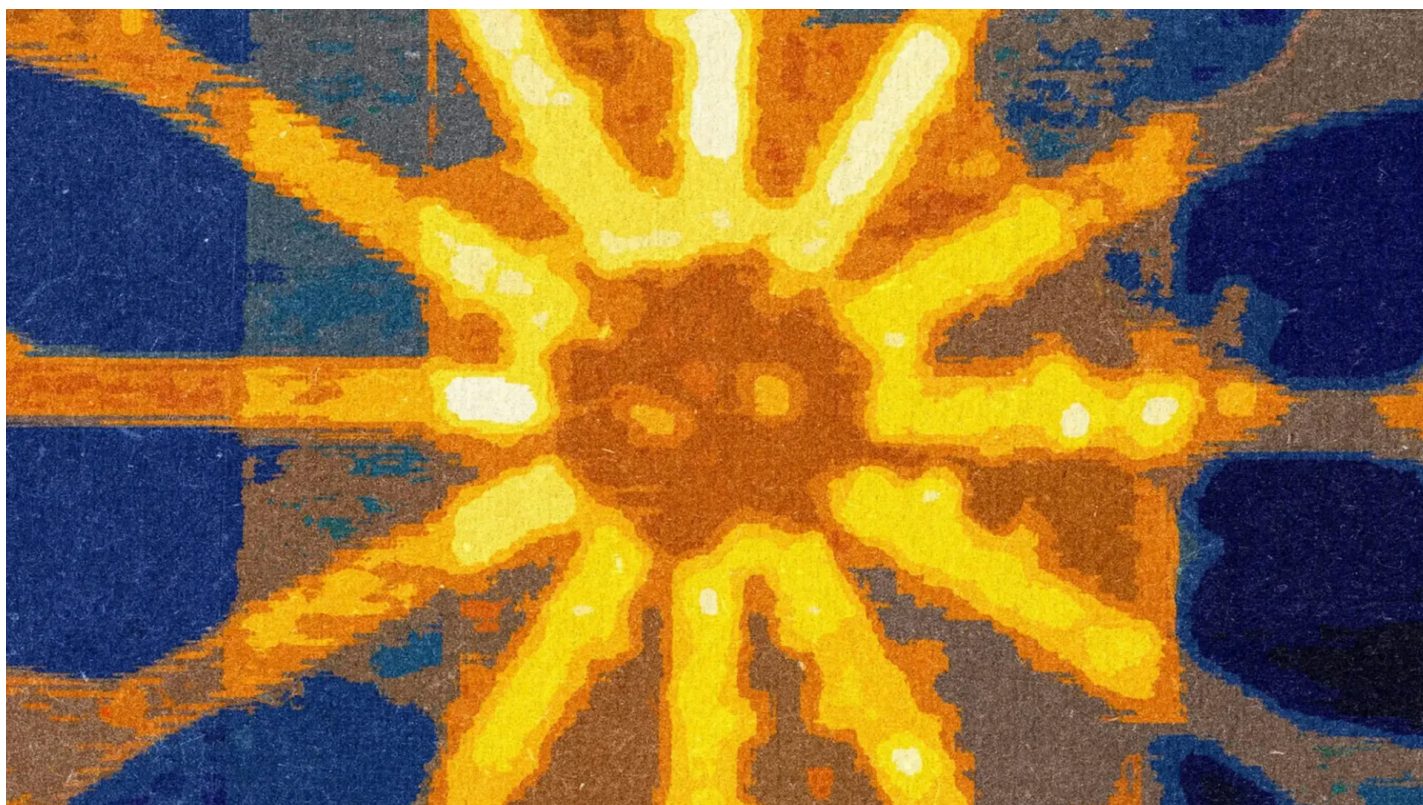
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Source: Carnegie Mellon University

Summary: Scientists at Carnegie Mellon University have discovered an unexpected form of the Hall effect, overturning the long-held assumption that this electrical response only appears when a magnetic field points perpendicular to a material. Beyond expanding a century-old principle of physics, the finding could eventually lead to simpler magnetic sensors for electronics, transportation, and medical technologies.

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FULL STORY



An atomic force microscopy scan of a nanometer-sized device developed by Carnegie Mellon scientists that demonstrates a new form of the Hall effect. Credit: Carnegie Mellon

Carnegie Mellon University researchers have identified an unusual magnetic response that overturns a long-standing assumption about the Hall effect, a foundational principle used to study how materials behave electrically and magnetically.

The findings, published in *Nature Materials*, expand scientists' understanding of the Hall effect and could eventually support simpler and more flexible magnetic sensors for electronics, transportation and medical imaging.

Rethinking a Century-Old Physics Effect

Scientists have relied on the Hall effect for more than a century. In 1879, Edwin Hall discovered that when a magnetic field is applied perpendicular to a material carrying an electric current, the moving charges are pushed to one side. This creates a voltage that researchers can measure.

That signal reveals important information about a material, including whether its current is carried by positive or negative charges, how many charge carriers are present, and how easily they can move. Hall effect sensors are now widely used in technologies ranging from cars to computer keyboards.

Researchers in Carnegie Mellon's Department of Physics, working in the Lab for Investigating Quantum Materials, Interfaces and Devices (LIQUID), have now demonstrated a different form of the effect.

"For a long time, people thought the Hall effect only worked when the magnetic field was applied perpendicular to the plane of the film. We've shown that that's not true -- you can also get a response when the field is in-plane," said Simranjeet Singh, an associate professor of physics.

The result shows that a Hall response tied to magnetization can occur in more than one direction. That gives physicists a new way to investigate multidimensional magnetic and topological structures in condensed matter systems.

"Beyond fundamental importance, this discovery can enable novel planar device architectures and sensor types, such as vector magnetometry, via measuring the out-of-plane and in-plane anomalous Hall effect signals in the same device," Singh said.

Turning a Prediction Into an Experiment

Scientists had previously predicted an in-plane anomalous Hall effect in theory, but no experiment had successfully demonstrated it before this work.

"People proposed it and ideas were out there, but it's very difficult to make a magnetic material with the right symmetry to do it," Singh said. "What we did was we found a material with the right symmetry, and we made it magnetic."

Creating the nanometer-sized devices required for the experiment involved expertise in two-dimensional quantum materials. Singh worked with Jyoti Katoch, an associate professor of physics who specializes in fabricating devices from such materials.

The research team, which included postdoctoral researchers I-Hsuan Kao and Ravi Kumar, began with tantalum iridium telluride (TaIrTe₄). Its crystal structure has the symmetry needed to support a multidimensional Hall effect. The researchers reduced the material to only a few atomic layers in thickness, then placed it next to a magnetic layer, Cr₂Ge₂Te₆ (CGT).

Because the two layers sit so closely together, magnetic behavior from the CGT influences the normally nonmagnetic TaIrTe₄. This gives the TaIrTe₄ magnetic properties while allowing it to retain its underlying electronic characteristics.

"This truly demonstrates the power of building atomically precise heterostructures of emergent two-dimensional quantum materials to obtain on-demand electronic and magnetic properties," Katoch said.

One Device, Multiple Magnetic Directions

Inside the atomically thin devices, the researchers detected both the familiar Hall signal and a second, unconventional signal associated with magnetization lying within the plane of the material.

That difference has a potentially important practical consequence. A single ultrathin device can detect magnetic fields along more than one axis.

"We have broadened the potential application of these materials," Singh said. "You can do multidimensional magnetic sensing with one sensor only. Before, you needed to put two sensors to measure the magnetic field in two directions."

The work therefore points toward magnetic sensing systems that could perform measurements in multiple directions without requiring separate sensors for each one.

Explaining the Unusual Hall Response

Alongside the experiments, Shubhayu Chatterjee, an assistant professor of physics, used theoretical modeling to investigate why the effect appears and how the symmetry of the combined materials makes it possible.

"We found that the reduced symmetry due to pairing with CGT allows additional spin-orbit coupling at the interface. These spin-orbit coupling terms are crucial for the in-plane anomalous Hall effect to emerge once CGT becomes ferromagnetic at low temperatures. While certain features of the observed anomalous Hall effect signal are consistent with an intrinsic origin, a detailed characterization of few-layered TaIrTe4 is needed to nail down the precise mechanism," Chatterjee said.

The LIQUID team is now investigating additional material combinations that could produce the same unconventional Hall response. The researchers are also testing how the device behaves at room temperature, which will be an important requirement if the technology is eventually used in practical applications.

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Materials provided by **Carnegie Mellon University**. Original written by Amy Pavlak Laird. *Note: Content may be edited for style and length.*

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